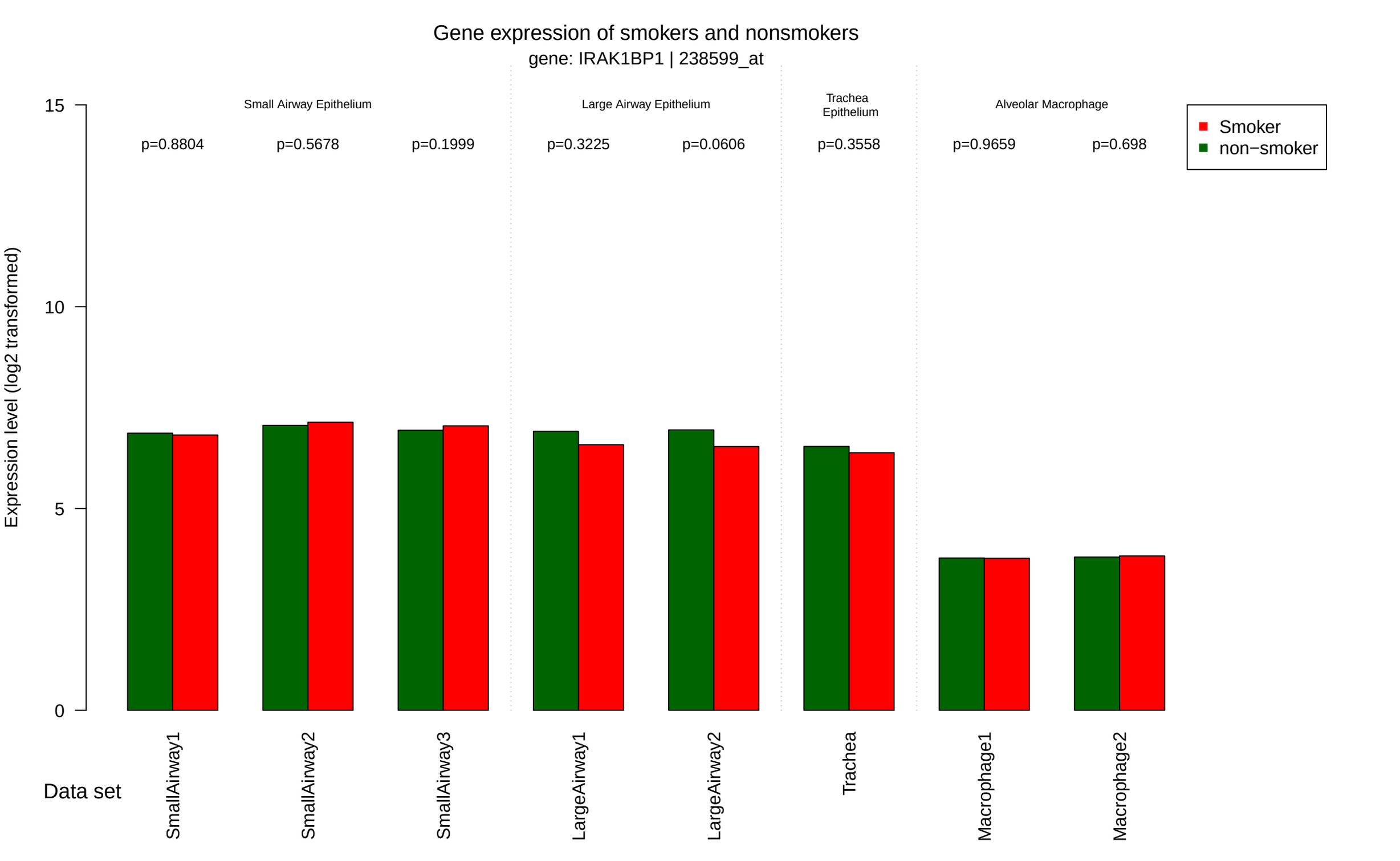
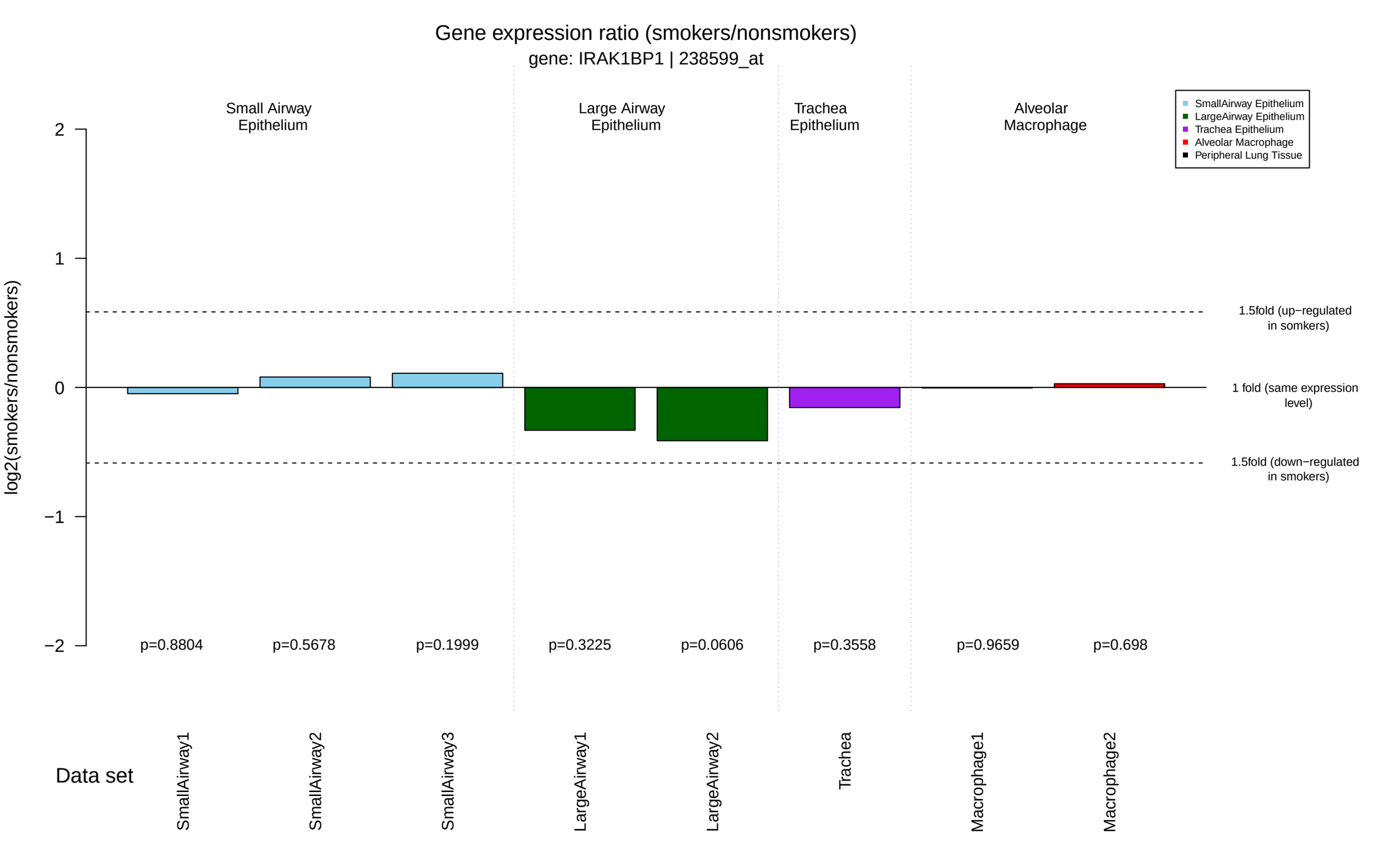
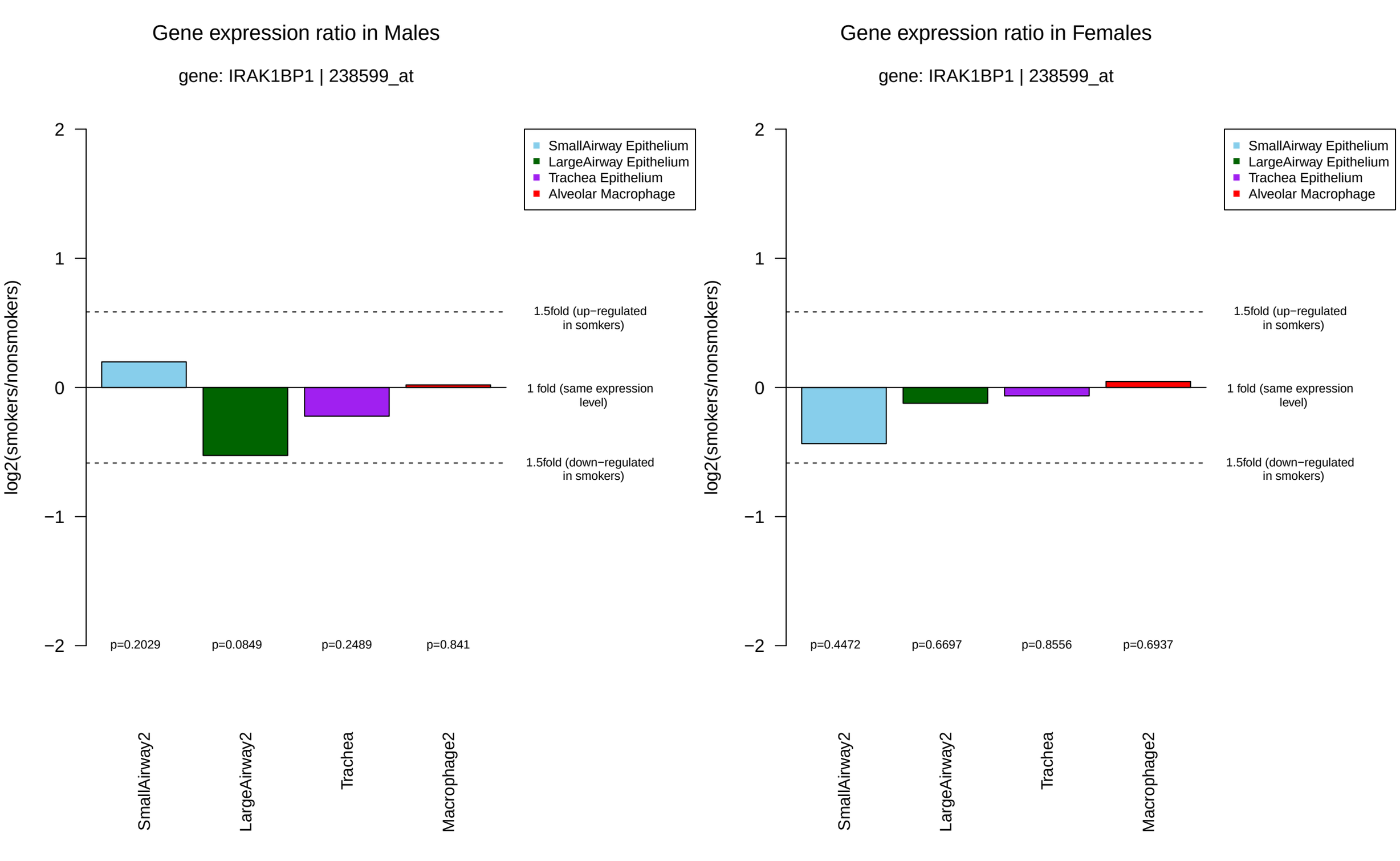
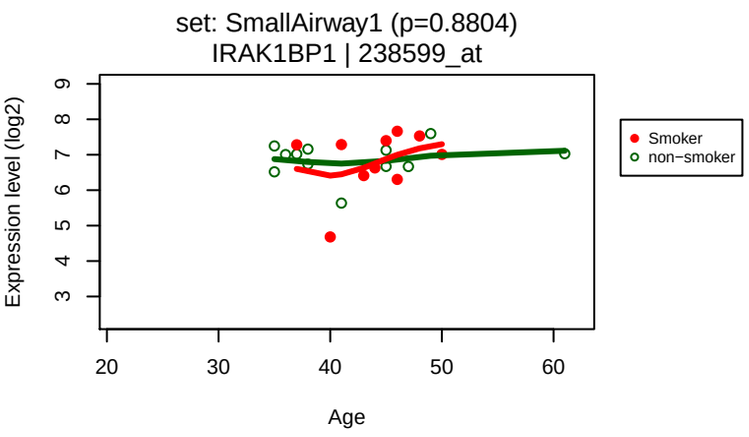
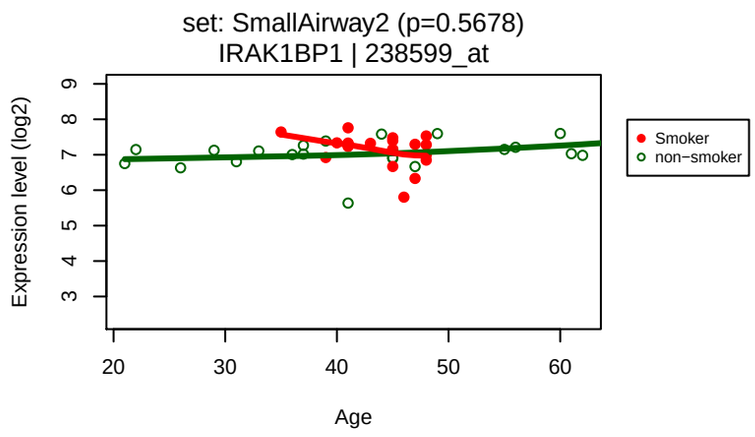
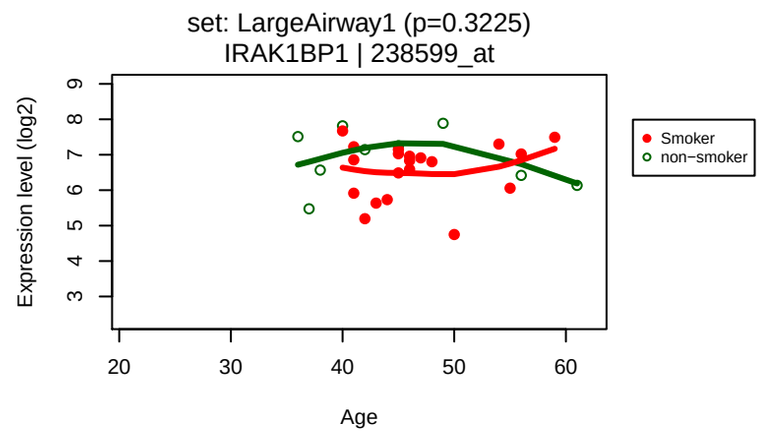

Gene expression of smokers and nonsmokers
gene: IRAK1BP1 | 238599_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8804
p=0.5678
p=0.1999
p=0.3225
p=0.0606
p=0.3558
p=0.9659
p=0.698
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

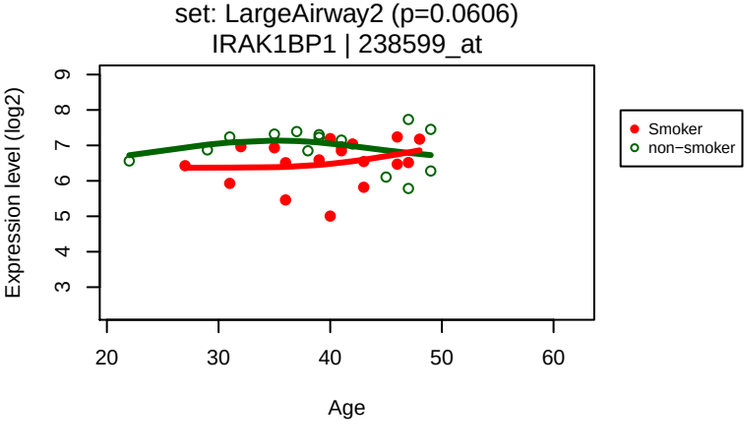

Gene expression ratio (smokers/nonsmokers)
gene: IRAK1BP1 | 238599_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8804
p=0.5678
p=0.1999
p=0.3225
p=0.0606
p=0.3558
p=0.9659
p=0.698
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

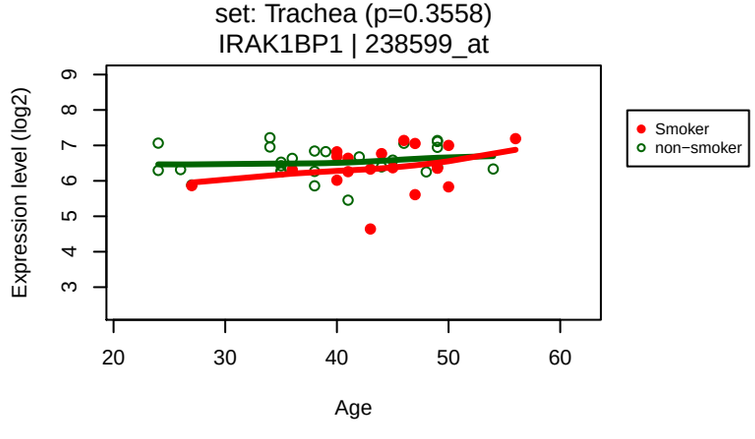

Gene expression ratio in Males
gene: IRAK1BP1 | 238599_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2029
p=0.0849
p=0.2489
p=0.841
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRAK1BP1 | 238599_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4472
p=0.6697
p=0.8556
p=0.6937
SmallAirway2
LargeAirway2
Trachea
Macrophage2

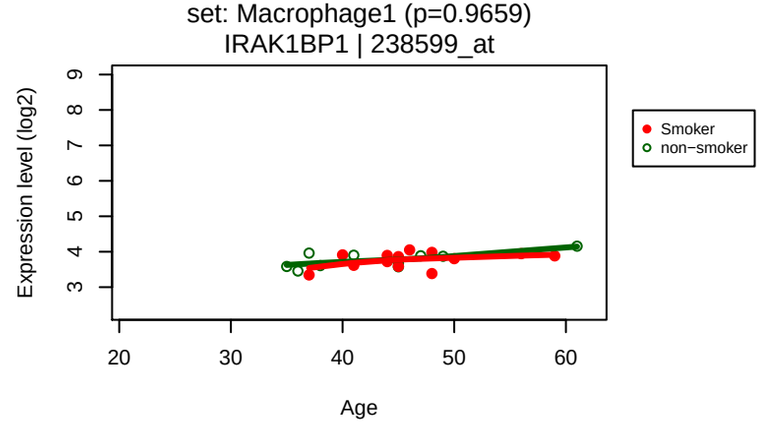

set: SmallAirway1 (p=0.8804)
IRAK1BP1 | 238599_at
Expression level (log2)
Age
Smoker
non−smoker

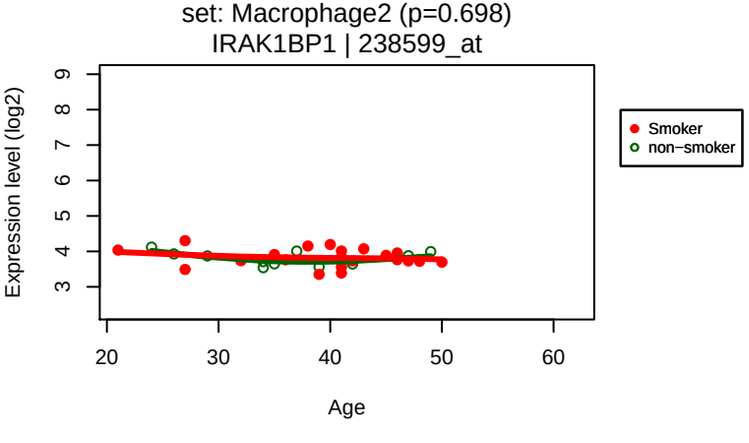

set: SmallAirway2 (p=0.5678)
IRAK1BP1 | 238599_at
Expression level (log2)
Age
Smoker
non−smoker

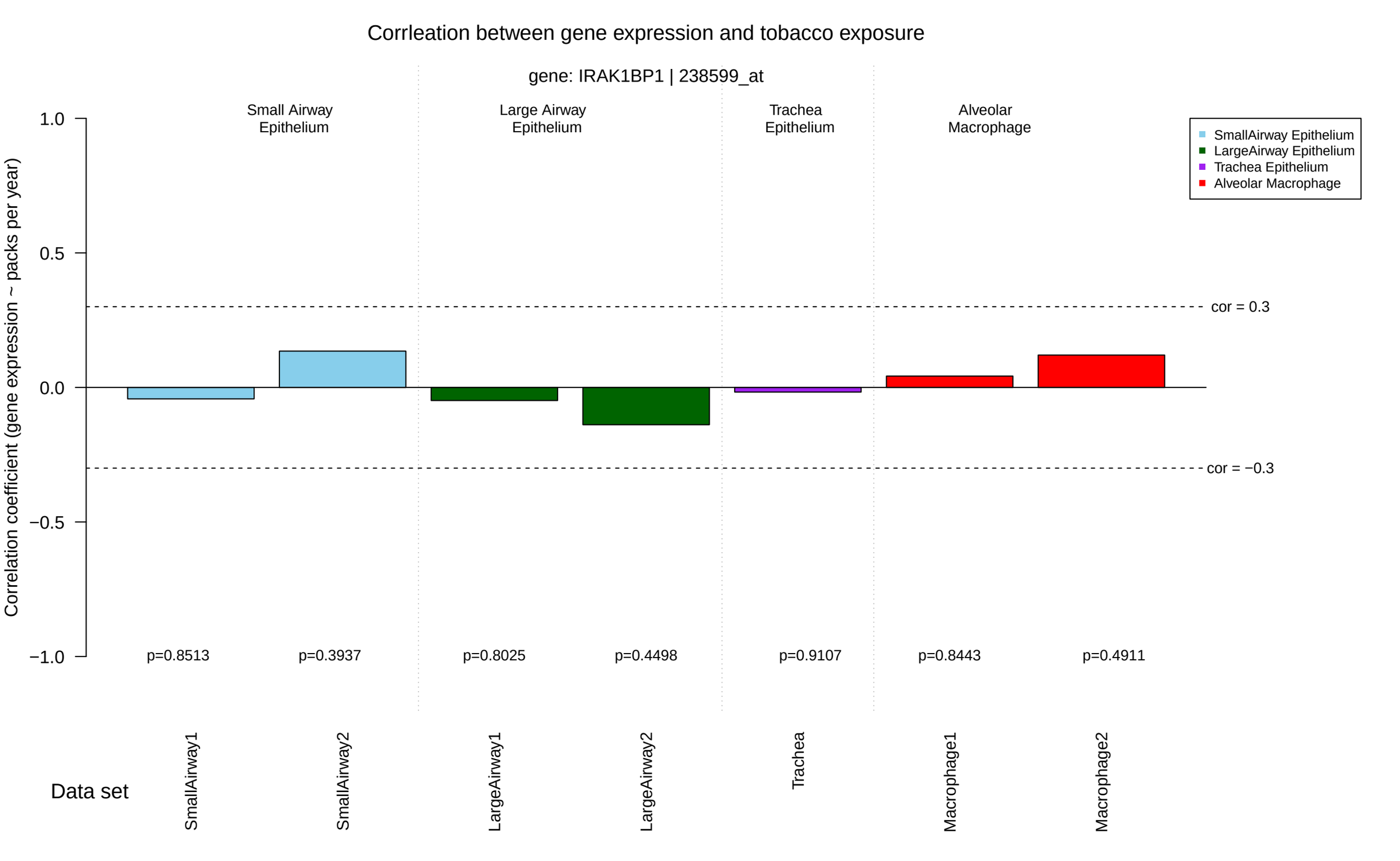

set: LargeAirway1 (p=0.3225)
IRAK1BP1 | 238599_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0606)
IRAK1BP1 | 238599_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3558)
IRAK1BP1 | 238599_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9659)
IRAK1BP1 | 238599_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.698)
IRAK1BP1 | 238599_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRAK1BP1 | 238599_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8513
p=0.3937
p=0.8025
p=0.4498
p=0.9107
p=0.8443
p=0.4911
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2